Mid-term Examination B卷

PHYSICS · Paper 4 (A2)

Centre No.: _____ **Class:** _____

Candidate No.: _____ **Candidate name:** _____

Exam Duration: 60 mins

Additional Materials: Answer Booklet/Paper

Graph Paper · List of Formulae

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, Candidate number and name on all the work you hand in.

Write in dark blue or black pen. You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer all questions.

Give non-exact numerical answers correct to 3 significant figures, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 100.

Topics: Motion in a circle · Gravitational fields · Oscillations

5 questions · 100 marks

Question 1 · Motion in a circle · 20 marks

Conical pendulum on a spring — tension, spring constant & circular motion

2 A steel sphere of mass 0.29 kg is suspended in equilibrium from a vertical spring. The centre of the sphere is 8.5 cm from the top of the spring, as shown in Fig. 2.1.

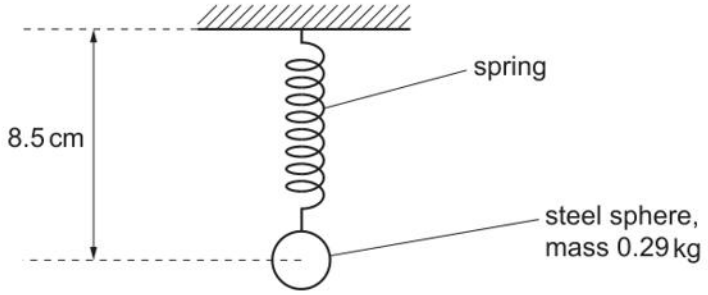


Fig. 2.1

The sphere is now set in motion so that it is moving in a horizontal circle at constant speed, as shown in Fig. 2.2.

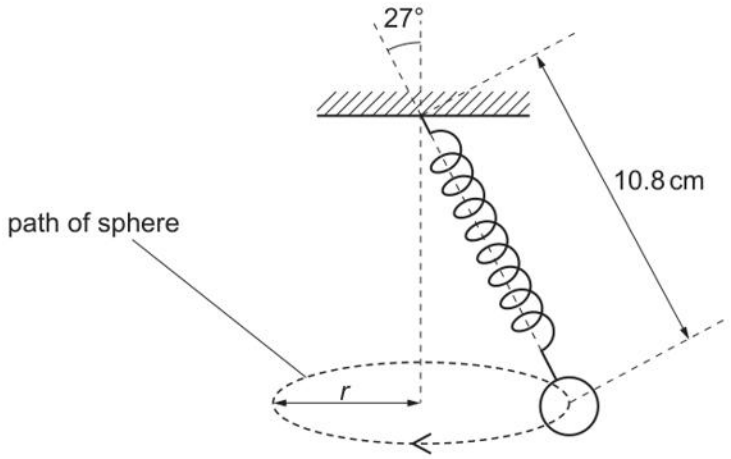


Fig. 2.2

The distance from the centre of the sphere to the top of the spring is now 10.8 cm.

(a) Explain, with reference to the forces acting on the sphere, why the length of the spring in Fig. 2.2 is greater than in Fig. 2.1.

.....

.....

.....

.....

..... [5]

(b) The angle between the linear axis of the spring and the vertical is 27°.

(i) Show that the radius r of the circle is 4.9 cm.

[2]

(ii) Show that the tension in the spring is 3.2 N.

[3]

(iii) The spring obeys Hooke's law.

Calculate the spring constant, in N cm^{-1}, of the spring.

spring constant = ... N cm^{-1} [3]

(c) **(i)** Use the information in **(b)** to determine the centripetal acceleration of the sphere.

centripetal acceleration = ... m s^{-2} [3]

(ii) Calculate the period of the circular motion of the sphere.

period = ... s [4]

[Total: 20]

Question 2 · Gravitational fields · 20 marks

Gravitational field lines & circular orbits — planet mass from r^3–$1/\omega^2$ graph

1 **(a)** The point P in Fig. 1.1 represents a point mass.

On Fig. 1.1, draw lines to represent the gravitational field around P.

● P

Fig. 1.1

[4]

(b) A moon is in circular orbit around a planet.

Explain why the path of the moon is circular.

.....

.....

.....

..... [4]

(c) Many moons are in circular orbit about a planet.

The angular velocity of a moon is ω when the orbit of the moon has a radius r about the planet.

Fig. 1.2 shows the variation of r^3 with $1/\omega^2$ for these moons.

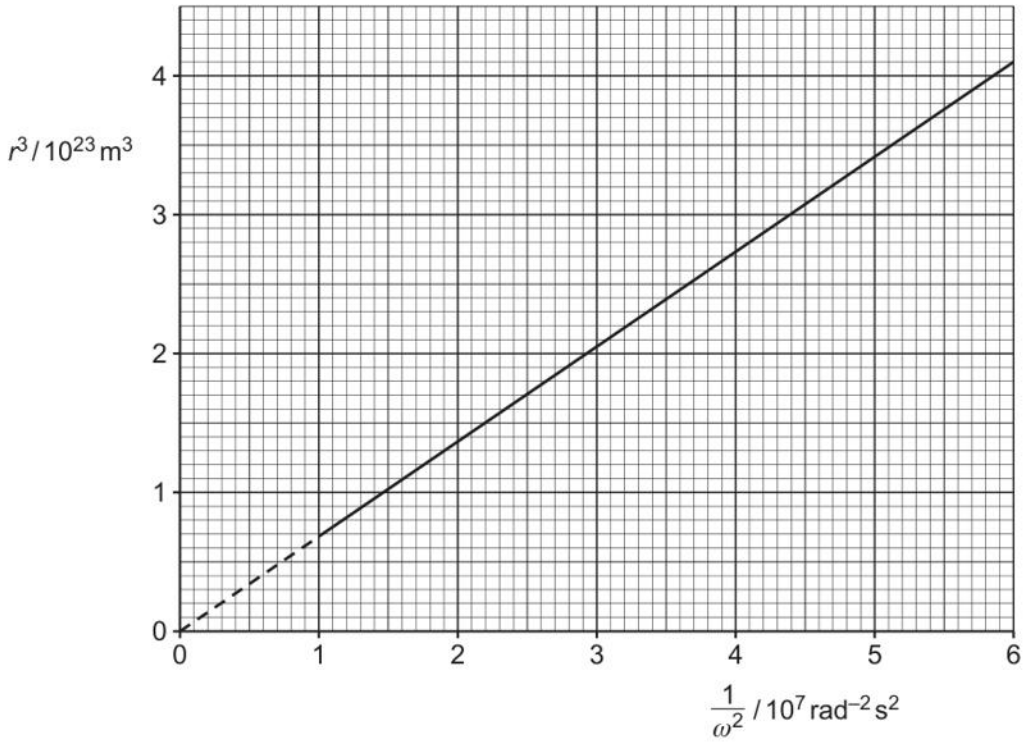


Fig. 1.2

(i) Show that the mass M of the planet is given by the expression

$$M = \frac{\text{gradient}}{G}$$

where G is the gravitational constant.

[4]

(ii) Use Fig. 1.2 and the expression in **(c)(i)** to show that the mass M of the planet is 1.0×10^{26} kg.

[2]

(iii) Determine the speed of a moon in orbit around the planet with an orbital radius of 1.2×10^{8} m.

speed = ... $m\,s^{-1}$ [6]

[Total: 20]

Question 3 · Gravitational fields · 20 marks

Newton's law of gravitation — satellite orbit relation, geostationary height & angular speed

1 **(a)** State Newton's law of gravitation.

.....

.....

..... [4]

(b) A satellite is in a circular orbit around a planet. The radius of the orbit is R and the period of the orbit is T. The planet is a uniform sphere.

Use Newton's law of gravitation to show that R and T are related by

$$4\pi^2R^3 = GMT^2$$

where M is the mass of the planet and G is the gravitational constant.

[4]

(c) The Earth may be considered to be a uniform sphere of mass 5.98×10^{24} kg and radius 6.37×10^{6} m.

A geostationary satellite is in orbit around the Earth.

Use the expression in **(b)** to determine the height of the satellite above the Earth's surface.

height = ... m [5]

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(d) Another satellite is in a circular orbit around the Earth with the same orbital radius and period as the satellite in **(c)**.

(i) Calculate the angular speed of the satellite in this orbit. Give a unit with your answer.

angular speed = ... unit ..... [3]

(ii) Despite having the same orbital period, the orbit of this satellite is not geostationary.

Suggest **two** ways in which the orbit of this satellite could be different from the orbit of the satellite in **(c)**.

1 .....

.....

2 .....

.....

[4]

[Total: 20]

Question 4 · Oscillations · 20 marks

Oscillating object — amplitude, angular frequency, total energy & KE graph

4 Fig. 4.1 shows the variation with time t of the height h above the ground of an object of mass 36 kg that is undergoing vertical simple harmonic motion.

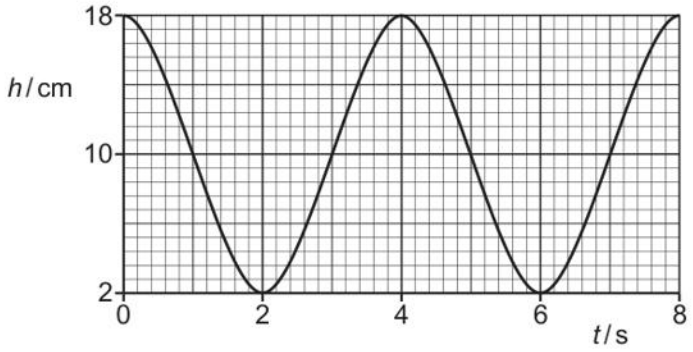


Fig. 4.1

(a) For the oscillations of the object:

(i) determine the amplitude x_0, in cm

x_0 = ... cm [2]

(ii) show that the angular frequency ω is 1.6 rad s^{-1}

[4]

(iii) determine the total energy E.

E = ... J [6]

(b) On Fig. 4.2, sketch the variation with h of the kinetic energy E_K of the object.

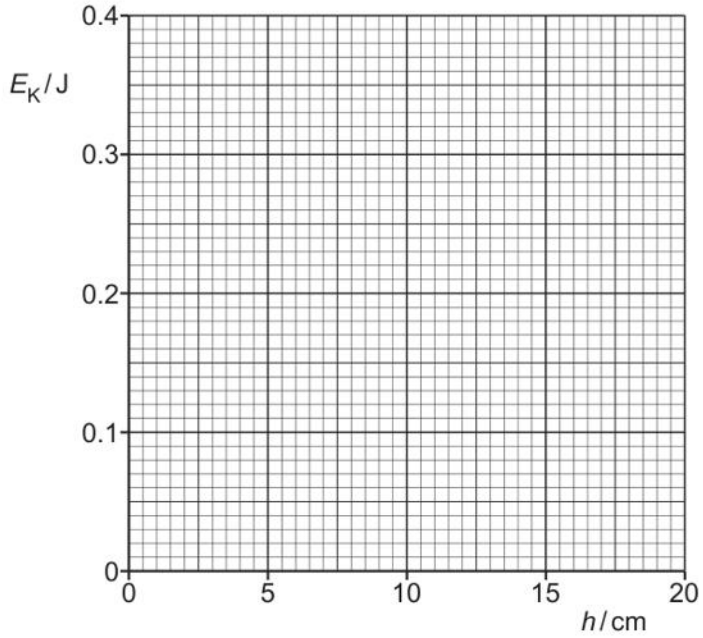


Fig. 4.2

[8]

[Total: 20]

Question 5 · Oscillations · 20 marks

Electrons oscillating in a metal rod — SHM, drift speed & alternating current

4 An electron in a metal rod moves randomly about a mean position. When an alternating voltage is applied to the ends of the rod, the mean position can be considered to oscillate with simple harmonic motion along the axis of the rod. Fig. 4.1 shows the variation with time t of the displacement x of the mean position from a fixed point on the axis of the rod.

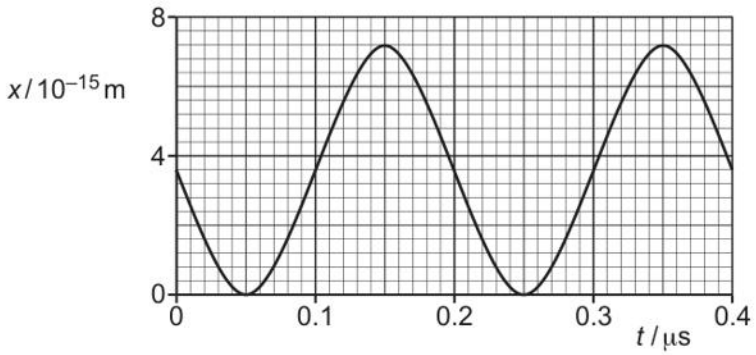


Fig. 4.1

(a) (i) Determine the amplitude of the oscillations.

amplitude = ... m [2]

(ii) Determine the angular frequency of the oscillations.

angular frequency = ... rad s^{-1} [2]

(iii) Use your answers in **(a)(i)** and **(a)(ii)** to show that the maximum drift speed v_0 of the electron is 1.1×10^{-7} m s^{-1}.

[4]

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(b) The rod has a cross-sectional area of $4.3\,cm^{-2}$ and contains a number density of conduction electrons (charge carriers) of $8.5 \times 10^{28}\,m^{-3}$.

All of the conduction electrons in the rod may be assumed to be oscillating in phase with, and with the same amplitude as, the oscillation shown in Fig. 4.1.

(i) Use the information in **(a)(iii)** to calculate the magnitude I_0 of the maximum current in the rod.

I_0 = ...A [3]

(ii) On Fig. 4.2, sketch the variation of the current I in the rod with time t between $t = 0$ and $t = 0.40\,\mu s$.

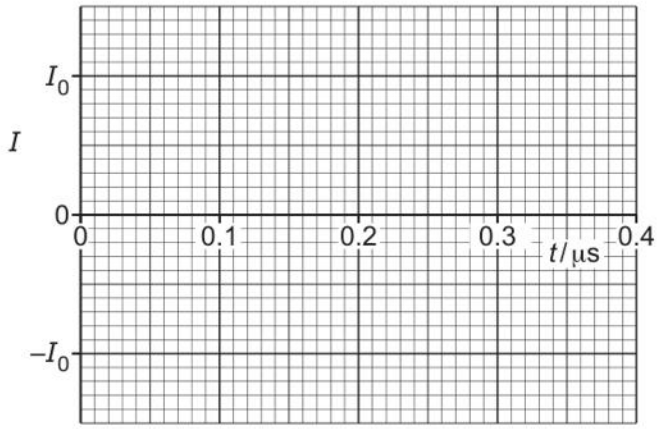


Fig. 4.2

[3]

(iii) Use your answers in **(a)(ii)** and **(b)(i)** to determine an expression for I in terms of t, where I is in A and t is in s.

I = ... [4]

(iv) Determine the root-mean-square (r.m.s.) current in the rod.

r.m.s. current = ...A [2]

[Total: 20]